College of Business
香港城市大學
City University of Hong Kong

FinTech and AI in Finance

Class 7: Bitcoin and Ethereum

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City University of Hong Kong

Bitcoin, Wallets, and Mining

Why Do People Hold or Use Bitcoin?

Different users want different things from the same asset.

- **Investment.** Some investors seek exposure to a scarce digital asset and accept large price fluctuations.
- **Control.** Self-custody lets holders authorize transfers themselves, with responsibility for protecting their keys.
- **Transfer.** Users can send BTC without a bank account on the network. Fees, confirmation time, and price risk affect its usefulness.

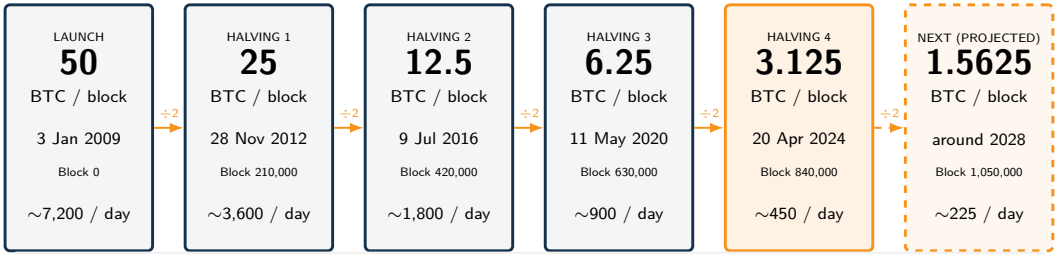
Compare the service with the user's alternatives, including a bank payment or regulated investment account.

Bitcoin Supply: A Predetermined Issuance Rule

- Under the current rules, the total supply approaches **21 million BTC**. New coins enter through the block subsidy.
- The subsidy halves every **210,000 blocks**, approximately every four years. Miners also earn transaction fees.
- The final fractions are expected around **2140**. Scarcity limits supply; it does not guarantee demand or a price increase.

Source: Bitcoin.org, issuance and supply.

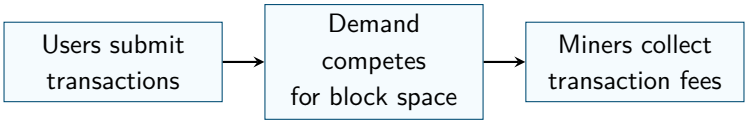
Bitcoin's Issuance Schedule: 2009 to the Next Halving



The rule is triggered by **block height**, not a calendar date. Daily issuance assumes the 10-minute target; actual block timing varies.

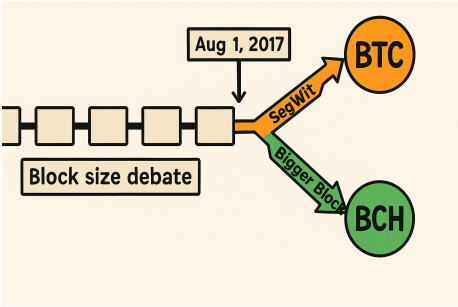
Sources: Bitcoin Core, mainnet consensus parameters; Blockstream block records at heights 210,000–840,000. The next date is an estimate; height and subsidy follow the protocol schedule.

Limited Block Space Creates a Fee Market



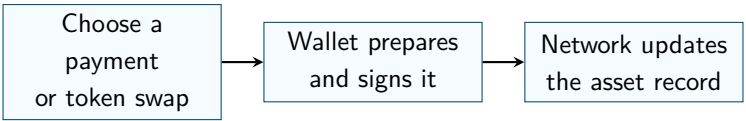
- A block can include only a limited amount of transaction data. During congestion, users may pay more or wait longer.
- A fee buys transaction processing, not a percentage ownership claim on the network.
- Halving reduces the subsidy; fee revenue also depends on demand for transactions.

The 2017 Fork Created BTC and BCH



- On **1 August 2017**, disagreement over block capacity produced two networks.
- **BTC** kept more conservative base-layer capacity; **BCH** increased it to process more transactions on-chain.
- They shared the earlier transaction history but then followed different rules and developed separate prices.

Ledger: Why Pay for a Hardware Wallet?



- Crypto owners want direct access without relying on an exchange to approve withdrawals. A familiar bank-style login leaves custody with an intermediary.
- Ledger sells hardware for self-custody. Assets stay on the blockchain, while the device protects the private keys used to authorize transfers.
- The customer takes responsibility for recovery. An address receives assets, while a private key or recovery phrase controls access.

Source: Ethereum.org, wallets.

Ledger: Security through Separate Signing

Custodial account: Coinbase.com

The provider controls the keys and manages account access.

Password recovery is possible, but customers depend on the provider's custody and operations.

Source: Ethereum.org, account and wallet security.

Self-custody: Ledger

The user controls keys held on a hardware device.

The device reduces online key exposure. Lost backups or an approved malicious transaction can still cause losses.

Ledger: Devices, Services, and Customer Trust

- Ledger currently sells Nano, Flex, and Stax devices. Its companion app connects users to buying, swapping, and staking services from third parties.
- The business combines hardware sales with connected services. Blockchain transaction fees are a separate payment to the network.
- Customers must trust the device and understand what they sign. Protecting a key cannot undo an authorized scam transaction.

Sources: Ledger device range; companion app. Checked 13 October 2026.

Crypto Markets: A Downturn Followed by a Rebound

Price snapshot

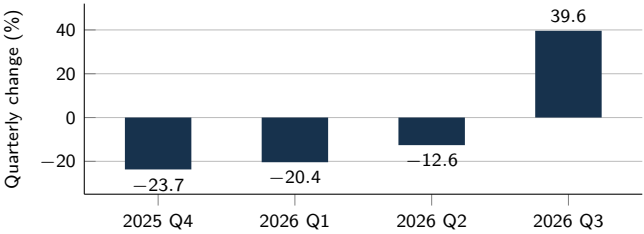
BTC: US$82,000

ETH: US$2,410

Approximate quotes observed on 20 October 2026; rounded, not daily closing prices.

Prices and market capitalization measure different things.

Total crypto market capitalization

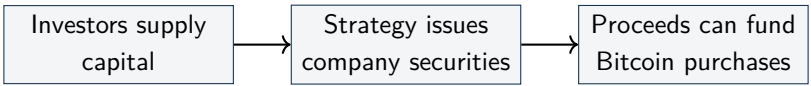


Three declining quarters were followed by a strong rebound. A rally does not automatically recover earlier losses.

Sources: CoinGecko BTC and ETH pages (20 Oct 2026); 2025 Annual, 2026 Q1, Q2, and Q3 reports. Bars are market-cap changes, not BTC/ETH returns.

Strategy: Financing a Bitcoin Treasury

Strategy (common-stock ticker: MSTR) combines a software business with a Bitcoin treasury strategy. Its securities give investors different claims on the company.



Financing	Investor claim and company cost
Common shares	Residual ownership; issuing more shares dilutes ownership percentages.
Preferred shares	Priority over common equity; dividend and redemption terms vary by series.
Debt	Repayment obligations; interest and conversion rights depend on the issue.

Buying MSTR common stock does not give a right to withdraw the company's BTC.

Sources: Strategy, company and securities information; 6 July 2026 disclosure. Simplified financing map.

Strategy: Asset Value and Cash Needs

NAV (net asset value) attributable to common shareholders: asset value less liabilities and preferred-share claims. NAV per share divides this value by common shares outstanding.

When funding is available

- A higher BTC price raises the value of the treasury.
- Issuing above NAV can raise NAV per share after costs.
- Debt and preferred shares can fund more BTC, but add claims ahead of common equity.

When the market turns down

- BTC value and the stock's premium to NAV can fall together.
- New financing may become more costly or harder to obtain.
- Cash needs require reserves, operating cash flow, financing or asset sales.

Bitcoin holdings alone produce no interest or dividends. An unrealized gain is not cash available to pay financing costs.

Dated example: on 6 July 2026, Strategy reported selling 3,588 BTC to fund preferred-stock dividends, with 843,775 BTC remaining.

Source: Strategy disclosure, 6 July 2026.

Strategy Case: Bitcoin Is Flat, but the Shares Fall

Hypothetical holding company: Bitcoin holdings stay at $100m, with no other assets or liabilities.



$$\text{Shareholder return} = \frac{100}{150} - 1 = \boxed{-33.3\%}$$

The initial 50% premium disappears even though Bitcoin's price is unchanged.

Worked Example: Issuing Shares Above Asset Value

Illustrative company: $100m of Bitcoin, 10m shares, and no liabilities. Net asset value (NAV) is $10 per share.



$$\text{New NAV per share} = \frac{\$115m}{11m} = \boxed{\$10.45}$$

Issuing above NAV increases assets per share in this example; issuing below NAV would reduce them.

Assumes no issue costs and no change in Bitcoin's price.

Worked Example: Mining Revenue and Costs

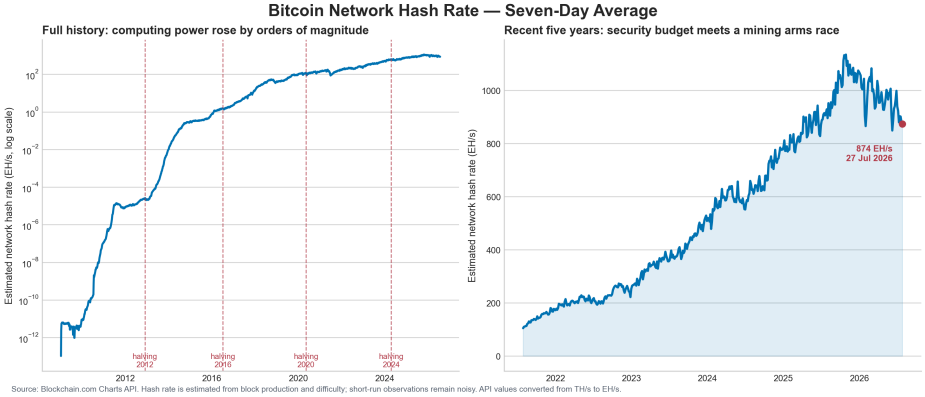
Illustrative month: A miner maintains 1% of the network's computing power. Total network rewards and transaction fees are worth US$10 million. The miner's monthly operating costs are US$120,000.

$$\text{Expected revenue} = 1\% \times \text{US\$10 million} = \text{US\$100,000}$$

$$\text{Expected operating profit} = 100,000 - 120,000 = \boxed{-\text{US\$20,000}}$$

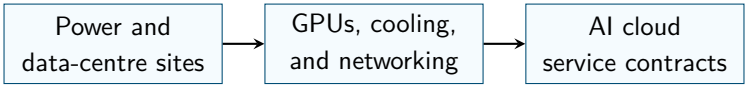
- A 1% share of computing power gives roughly a 1% expected share of rewards. Actual rewards fluctuate.
- Miners compete on costs as well as computing power. Electricity and hardware expenses can exceed mining revenue.

Competition in Bitcoin Mining



A miner must add computing capacity just to maintain its revenue share when the network expands. More machines do not guarantee higher profit.

IREN: From Bitcoin Mining to AI Infrastructure



- In November 2025, IREN announced a **five-year, US$9.7bn** AI infrastructure agreement with Microsoft.
- Power and sites support the transition, but AI workloads require substantial new equipment and capital.
- The business shifts from variable Bitcoin rewards toward contracted computing services, with delivery and customer-concentration risks.

Source: IREN, 3 November 2025; contract value, not realized revenue.

Ethereum and Smart Contracts

Bitcoin and Ethereum Serve Different Purposes

Bitcoin

A scarce digital asset and payment network.

Proof of work: miners commit computing resources.

Predetermined issuance approaching 21m BTC.

Source: Ethereum.org, introduction.

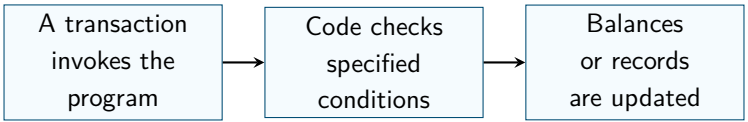
Ethereum

A programmable network for assets and financial applications.

Proof of stake: validators commit ETH.

ETH pays for computation and supports network security.

A Smart Contract Executes a Program on the Ledger



- The program can hold assets and apply rules for transfers, lending, or trading.
- Execution follows the code and submitted inputs; it does not independently verify a delivery or a legal claim.
- Deploying code does not make it run continuously: transactions are needed to trigger actions.

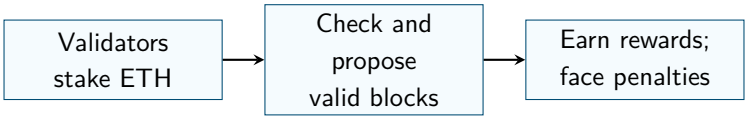
Source: Ethereum.org, smart contracts.

Conditional Supplier Payment

Escrow holds the buyer's funds until agreed conditions are met.



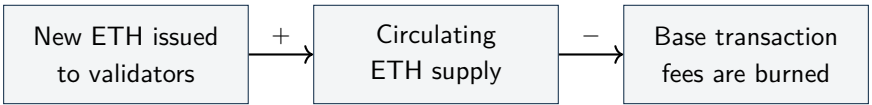
Staking: Commit Capital to Help Run the Network



- Validators put ETH at risk. Persistent failures or dishonest conduct can result in penalties.
- Investors can operate a validator or use a service, which adds provider fees and operational exposure.
- Rewards are paid in ETH; the dollar value still depends on the market price.

Source: Ethereum.org, proof of stake.

ETH Supply Reflects Both Issuance and Burning



$$\text{Change in supply} = \text{new issuance} - \text{ETH burned}$$

- Greater network use can increase fee burning, but ETH supply need not fall in every period.
- Transaction fees also include validator tips. A fee paid by a user is not all validator income.

Source: Ethereum.org, gas and fees.

Worked Example: Staking Yield and Dollar Return

Illustrative year: An investor stakes 10 ETH worth US$2,000 each. Net staking rewards add 4% to the ETH balance, while the ETH price falls 20% to US$1,600.

$$\text{Ending value} = 10 \times 1.04 \times 1,600 = \text{US\$16,640}$$

$$\text{Dollar return} = \frac{16,640}{20,000} - 1 = \boxed{-16.8\%}$$

- Staking rewards compensate validators for helping verify transactions. A positive reward in ETH can accompany a loss in dollars.
- Validator penalties and withdrawal delays add separate risks.

Worked Example: A Network Fee in Dollars

Illustrative transfer: the quoted network fee is 0.001 ETH and ETH trades at \$2,000.

$$\text{Dollar fee} = 0.001 \times \$2,000 = \boxed{\$2}$$



The same \$2 fee is 10% of a \$20 payment, but only 0.4% of a \$500 payment.

Multiple-Choice Question: A Bitcoin Holding Company's Premium

A US-listed company holds Bitcoin worth $100 million. Its shares have a total market value of $150 million. Assume it has no other assets or liabilities.

What premium do shareholders pay relative to the value of its Bitcoin holdings?

A) 33.3%

B) 50%

C) 150%

D) 0%

Multiple-Choice Question: A Halving and Miner Revenue

Before a hypothetical halving, block subsidies provide 80% of a miner's revenue and transaction fees provide 20%. The subsidy halves. The coin price, fee revenue, and number of blocks mined stay unchanged.

By what percentage does total revenue fall?

A) 20%

B) 40%

C) 50%

D) 80%

Multiple-Choice Question: Staking Income and Investment Risk

An investor receives staking rewards in Ether but measures investment performance in US dollars. The number of Ether held increases during the year.

Which statement about the investment is correct?

A) More Ether guarantees a positive dollar return.

B) Staking rewards compensate the investor for any fall in Ether's price.

C) Reinvesting rewards removes exposure to the token's market price.

D) The dollar value can fall even though the investor holds more Ether.

A Smart-Contract Auction

The seller deposits a unique digital collectible (an NFT) into a contract. Assume no administrator can bypass the auction rules.

Bid sequence	USDC offered	Contract response
Alice	100	Holds her bid
Bob	120	Leads. Alice may withdraw 100
Charlie	110	Rejects the lower bid

After the deadline, a settlement transaction transfers the NFT to Bob and 120 USDC to the seller.

Could the seller collect Bob's payment but refuse to deliver the token?

Hypothetical contract, ignoring fees. NFT ownership alone grants no copyright.

Building a Protocol and Building a Financial Company



Vitalik Buterin 维塔利克 · 布特林

Proposed Ethereum in 2013 as a general programmable blockchain.

Developers build applications on shared infrastructure.



Brian Armstrong

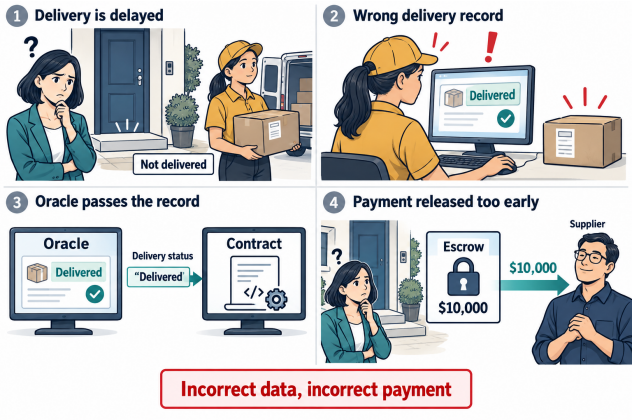
Co-founded Coinbase in 2012 to make crypto access easier.

The company earns service fees and is responsible for custody and operations.

Sources: Ethereum and Coinbase histories. Portraits are editorial illustrations, not photographs.

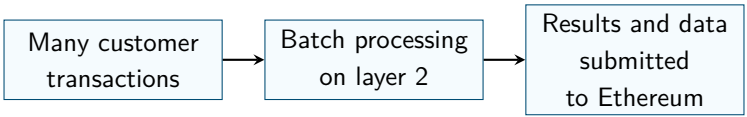
Delivery Data and the Oracle

An **oracle** supplies outside information to a smart contract.



Source: Ethereum.org, smart contract security.

Rollups: Share Settlement Costs across Transactions



- Layer 2 adds processing capacity while using Ethereum for settlement and security.
- Batching can lower the cost per transaction. Users still pay for processing and data.
- Deposits, withdrawals, and network availability depend on the service used; cheaper does not mean identical.

Source: Ethereum.org, scaling.

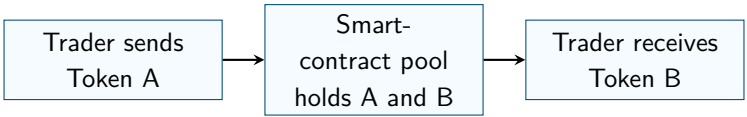
A Token Sale Is Not Necessarily a Share Issue

What is sold?	What must the buyer establish?
Equity	Ownership and contractual rights to the company.
Service token	What service it buys, who provides it, and on what terms.
Governance token	What decisions holders can influence and whether any cash-flow rights exist.

An initial coin offering (ICO) raises funds by selling tokens. The label alone does not establish ownership, income rights, or regulatory permission.

Trading Platforms and Market Access

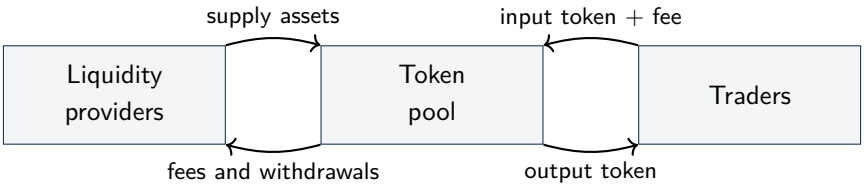
Uniswap: Trading against a Liquidity Pool



- A **liquidity pool** holds assets supplied by investors. The trader exchanges against these reserves.
- An **automated market maker (AMM)** uses a pricing rule instead of matching two posted orders.
- Liquidity providers earn fees but bear the risk of changes in the pool's asset values.

Source: Uniswap documentation.

Who Supplies Capital and Who Pays Fees?



Developers and interfaces provide access to the pool. Protocol fees and interface fees depend on the version and service.

AMM Pricing: Buying 10 ETH

Let x be ETH reserves and y USDC reserves. Initially, $x = 100$ and $y = 200,000$, so the marginal price is 2,000 USDC per ETH.

$$xy = k = 100 \times 200,000 = 20,000,000$$

Buying 10 ETH leaves 90 ETH in the pool:

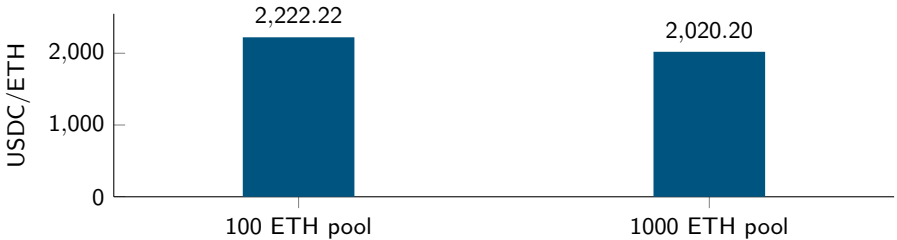
$$y_{\text{after}} = 20,000,000/90 = 222,222.22,$$
$$\text{USDC paid} = 222,222.22 - 200,000 = 22,222.22.$$

Average execution price is **2,222.22 USDC per ETH**. The final marginal price is $222,222.22/90 = 2,469.14$.

Hypothetical Uniswap v2-style full-range pool. No trading fees, gas, or LP deposits/withdrawals during the trade. Source: Uniswap documentation.

The Same Trade in a Deeper Pool

Average execution price for buying 10 ETH. Initial price: 2,000 USDC/ETH.



Deep pool: 1,000 ETH and 2m USDC. Payment: **20,202.02 USDC**.

Outside price: 2,000. Would arbitrageurs buy ETH from or sell ETH to the shallow pool?

Worked Example: Liquidity-Provider Fee Income

Illustrative day: trading volume is \$2m, the fee is 0.3%, and an investor provides 5% of the pool.

$$\text{Total fees} = \$2m \times 0.3\% = \$6,000$$

$$\text{Investor's fee share} = 5\% \times \$6,000 = \boxed{\$300}$$

- This is fee income, not total investment profit. The pool's assets can gain or lose value.
- Assume the fee is fully allocated to liquidity providers and the investor's share stays constant.

Hyperliquid: A Small Team with a Large Trading Business

Core team

11 people

Founder-reported headcount in October 2025.

Revenue scale

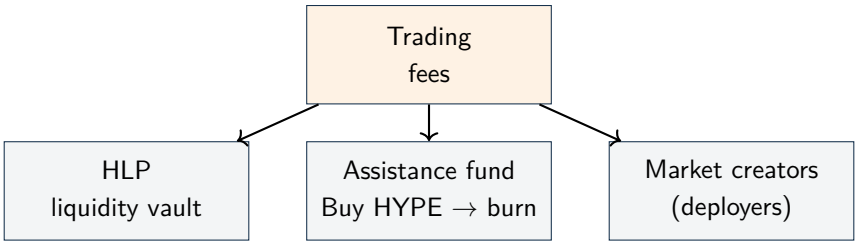
$1bn+

Founder-described annual revenue scale in the same interview.

- The main product is on-chain perpetual futures: leveraged contracts with no expiry date.
- Software can serve high trading volume with a small core team. Validators, liquidity providers, and external builders also support the business.
- The interview's revenue claim is not audited corporate net income.

Source: Jeff Yan, TBPN interview, 22 October 2025.

Hyperliquid: Where Trading Fees Go

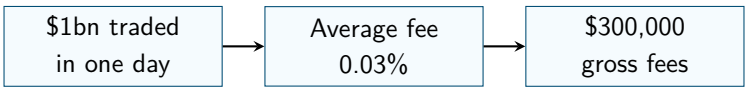


- The assistance fund converts fees into HYPE; these tokens are burned, reducing supply.
- Protocol fee flows are not the same as revenue available to a company's shareholders.

Source: Hyperliquid, fee documentation, checked 20 October 2026.

Worked Example: Trading Volume Is Not Profit

Illustrative exchange, not Hyperliquid accounts:



Contribution after rebates $= \$300,000 - \$60,000 = \$240,000$

- Technology, staff, compliance, and other operating costs must still be paid.
- To compare businesses, distinguish trading volume, gross fees, retained revenue, and profit.

Short-Answer Question: Automated Payment and Incorrect Data

A US importer uses an app that pays a supplier automatically when a delivery database reports that goods have arrived. An incorrect entry triggers payment before delivery.

1. Explain why correct software execution did not protect the importer.
2. Identify one check that could reduce incorrect payments.
3. State one recovery or dispute procedure the business still needs.

HKSAR: Regulated Access to Virtual Assets

- Centralized trading platforms operating in HKSAR, or actively marketing there, require SFC licensing.
- Spot Bitcoin and Ether ETFs began trading on HKEX in April 2024, offering access through securities accounts.
- Licensing changes custody and conduct requirements. It does not remove crypto-asset price risk.

Source: SFC, platform requirements. HKEX, spot virtual-asset ETF launch, April 2024.

Three Ways to Obtain Ether Exposure

Self-custody	**Trading platform**	**Spot ETF**
Hold ETH directly	Custody account	Hold fund shares
Manage keys and on-chain transactions	Rely on provider access and controls	Use a broker; pay fund fees

The same underlying asset can be accessed through different claims, costs, and operating arrangements.

Worked Example: The Cost of ETF Access

Illustrative investment: \$10,000 in a spot crypto ETF. Assume a constant asset value, a 0.5% annual fund fee, and \$10 brokerage at purchase and sale.

$$\text{One-year fees} = 10,000 \times 0.5\% + 10 + 10 = \boxed{\$70}$$

$$\text{Fees as a share of investment} = 70/10,000 = 0.7\%$$

- The investor avoids managing private keys but pays for the fund and brokerage services.
- Bid–ask spreads and any premium or discount can add to the cost.

Stablecoins Account for Most Identified Illicit Flows

Chainalysis estimate for 2025: illicit addresses received at least $154bn.



These are shares of *identified illicit* volume, not the fraction of all stablecoin activity that is illicit.

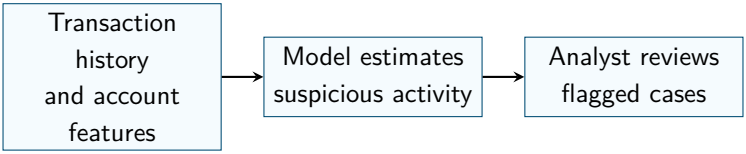
Source: Chainalysis, 2026 Crypto Crime Report.

A Trading Platform Is Also a Financial Intermediary

- **Custody:** safeguard client assets, control withdrawals, and reconcile customer balances.
- **Market conduct:** monitor manipulation, conflicts of interest, and asset-admission standards.
- **Financial crime:** identify customers and investigate suspicious transactions.

Source: SFC, requirements for licensed platforms.

An ML Application: Prioritizing Transaction Review



- Inputs can include transfer size, frequency, counterparties, and recent account changes.
- A model ranks cases for review. A high score does not by itself establish criminal conduct.
- As in Classes 3–5, evaluate performance on later data and examine which variables drive the score.

Illustrative workflow, not a claimed result for a named platform.

Chinese Mainland and HKSAR: Different Market Access

Chinese mainland

The **People's Bank of China (PBoC)**, China's central bank, and other agencies issued the February 2026 notice prohibiting domestic virtual-currency exchange and token-issuance financing.

The PBoC sets policy for money and payments. Business use of a shared ledger does not itself permit crypto trading.

HKSAR

Licensed platforms and authorized investment products provide regulated routes to access.

A HKSAR licence does not authorize the same business to serve Chinese mainland customers.

Sources: PBoC and CSRC, policy explanation, 6 February 2026; SFC platform licensing framework.

Worked Example: A Model Flags 100 Transfers

Illustrative review results: of 100 flagged transfers, investigators confirm 20 as suspicious; 80 are false alarms.

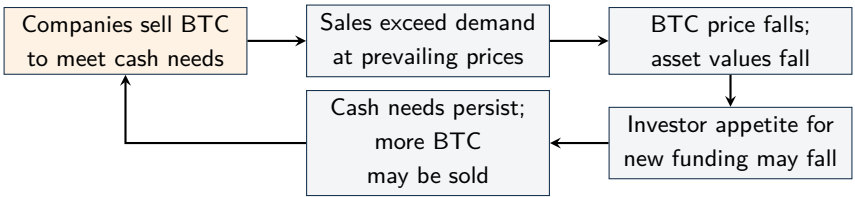


$$\text{Precision} = \frac{20}{100} = \boxed{20\%}$$

Review workload is high. To measure missed cases, we also need information about transfers the model did not flag.

Bitcoin Treasury Companies: Selling and Financing Risk

Conditional stress scenario: several treasury companies need cash while new financing is difficult and market liquidity is thin.



- **Amplifiers:** clustered maturities, low cash, simultaneous sales and thin markets. A shrinking share premium can further weaken financing; leveraged holders may also liquidate.
- **Buffers:** cash reserves, longer maturities, alternative funding, and buyers willing to absorb sales.
- **No automatic trigger:** a falling share price alone does not force a BTC sale. Obligations, collateral terms and management choices matter.

Sources: Wen (2025), treasury-company risks, working paper; Strategy sale disclosure (July 2026).

What Does a Crypto Downturn Reveal?

Participant	How a price decline can affect the business
Bitcoin miner	Dollar rewards fall if BTC earned is unchanged; power costs remain.
Bitcoin holding company	Asset value falls; financing obligations remain and the share premium may shrink.
ETH staker	More ETH can still be worth fewer dollars; staking is not price insurance.
Trading platform	Revenue depends on volume and fee rates. A sell-off can briefly increase trading.

Illustrative investment: start with \$100, lose 50%, then gain 50%.

$$100 \times (1 - 0.50) \times (1 + 0.50) = \boxed{75}$$

Still a **25% loss**. Recovering from a 50% loss requires a **100% gain**.

Discuss: which participants depend on token prices, and which depend on customer activity?